Chapter IV

Vinogradov's method in the theory of the Riemann zeta-function

§1. Vinogradov's mean value theorem

A mean value theorem is a bound for the number $J = J(P; n, k)$ of integer solutions of the following system of equations:

$$\begin{aligned} x_1 + \cdots + x_k &= y_1 + \cdots + y_k \\ x_1^2 + \cdots + x_k^2 &= y_1^2 + \cdots + y_k^2 \\ \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \\ x_1^n + \cdots + x_k^n &= y_1^n + \cdots + y_k^n \end{aligned} \tag{1}$$

$$1 \le x_1, \ldots, x_k, y_1, \ldots, y_k \le P.$$

Using the fact that

$$\int_0^1 \exp(2\pi i \alpha m) d\alpha = \begin{cases} 1, & \text{if } m = 0, \\ 0, & \text{if } m \neq 0, \ m \text{ an integer}, \end{cases}$$

we easily obtain the formula

$$J = J(P; n, k) = \int_0^1 \cdots \int_0^1 \left| \sum_{x \le P} \exp\bigl(2\pi i f(x)\bigr) \right|^{2k} d\alpha_1 \cdots d\alpha_n, \tag{2}$$

where $f(x) = \alpha_1 x + \cdots + \alpha_n x^n$. This formula says that $J = J(P; n, k)$ is the mean value of the $2k$-th power of the modulus of a trigonometric sum having an n-th degree polynomial in the exponent. We first prove four auxiliary results.

1. Lemma on the distribution of prime numbers

Lemma 1. *For any natural number n, if $x \ge (2n)^2$, then the interval $(x, 2x]$ contains at least n different prime numbers.*

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Proof. If $x \leq 1024$, the lemma can be checked directly; so we shall suppose that $x > 1024$.

The obvious inequalities

$$[x] \leq x < [x] + 1 < 2[x] \leq 2x < 2[x] + 2$$

imply that $(x, 2x] \subset ([x], 2[x] + 2)$, i.e.,

$$\psi(2x) - \psi(x) \leq \psi(2m+1) - \psi(m) \leq \psi(2m) - \psi(m) + \log(2m+1), \qquad m = [x]. \tag{3}$$

For any natural number m we have

$$\frac{1}{\sqrt{4m}} \leq \frac{1}{4^m}\binom{2m}{m} \leq \frac{1}{\sqrt{2m+2}}. \tag{4}$$

Furthermore, since

$$\log(k!) = \sum_{n \leq k} \log n = \sum_{n \leq k} \sum_{d \mid n} \Lambda(d) = \sum_{d \leq k} \Lambda(d) \sum_{n \leq k,\ n \equiv 0 \pmod{d}} 1 =$$
$$= \sum_{d \leq k} \Lambda(d) \sum_{rd \leq k} 1 = \sum_{r \leq k} \sum_{d \leq k/r} \Lambda(d) = \sum_{r \leq k} \psi(k/r),$$

it follows that

$$\log \binom{2m}{m} = \log \frac{(2m)!}{(m!)^2} = \sum_{r \leq 2m} \psi\left(\frac{2m}{r}\right) - 2\sum_{r \leq m} \psi\left(\frac{2m}{2r}\right) =$$
$$= \psi(2m) - \psi\left(\frac{2m}{2}\right) + \psi\left(\frac{2m}{3}\right) - \cdots + \psi\left(\frac{2m}{2m-1}\right).$$

Since $\psi(x)$ is a nondecreasing function, we have

$$\psi(2m) - \psi(m) \leq \log \binom{2m}{m} \leq \psi(2m) - \psi(m) + \psi(\tfrac{2}{3}m). \tag{5}$$

Consequently, from (4) and (5) we find that

$$\psi(2x) - \psi(x) \geq \log \binom{2m}{m} - \psi(\tfrac{2}{3}m) \geq m \log 4 - \frac{1}{2}\log 4m - \psi(\tfrac{2}{3}m), \tag{6}$$

where $m = [x]$. We use (5) to bound $\psi(x)$ from above. Setting $m_1 = [x/2]$, $m_2 = [x/4], \ldots$, and using (3), (4) and (5), we obtain

$$\psi(x) - \psi\left(\frac{x}{2}\right) \leq \log \binom{2m_1}{m_1} \leq m_1 \log 4 + \frac{1}{2}\log(2m_1 + 1),$$

$$\psi\left(\frac{x}{2}\right) - \psi\left(\frac{x}{4}\right) \leq m_2 \log 4 + \frac{1}{2}\log(2m_2 + 1),$$

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$$\psi\left(\frac{x}{4}\right) - \psi\left(\frac{x}{8}\right) \leq m_3 \log 4 + \frac{1}{2}\log(2m_3 + 1),$$

and so on. Summing both sides of these inequalities, we find that

$$\begin{aligned}\psi(x) &\leq (m_1 + m_2 + m_3 + \cdots)\log 4 + \\ &\quad + \frac{1}{2}\log(2m_1 + 1) + \frac{1}{2}\log(2m_2 + 1) + \frac{1}{2}\log(2m_3 + 1) + \cdots \\ &< \left(\frac{x}{2} + \frac{x}{4} + \frac{x}{8} + \cdots\right)\log 4 + \log^2(x + 1) \\ &\leq x \log 4 + \log^2(x + 1).\end{aligned}$$

From this and (6) we obtain (recall that $m = [x] \geq 1024$)

$$\begin{aligned}\psi(2x) - \psi(x) &\geq m \log 4 - \frac{1}{2}\log 4m - \frac{2}{3}m \log 4 - \log^2\left(\frac{2}{3}m + 1\right) \\ &\geq \frac{m}{3}\log 4 - \log^2 m.\end{aligned} \tag{7}$$

We note that

$$\psi(2m) - \psi(m) = \sum\nolimits_{m<n\leq 2m} \Lambda(n) = \sum\nolimits^{(1)}_{m<n\leq 2m} \Lambda(n) + \sum\nolimits^{(2)}_{m<n\leq 2m} \Lambda(n),$$

where $\sum^{(1)}$ denotes the sum over prime values of n and $\sum^{(2)}$ denotes the sum over composite n. To each n in the second sum for which $\Lambda(n) \neq 0$ we associate the prime p_n for which $p_n | n$. Obviously,

$$\Lambda(p_n) = \Lambda(n) \qquad \text{and} \qquad p_n \leq \sqrt{2m}.$$

Hence,

$$\sum\nolimits^{(2)}_{m<n\leq 2m} \Lambda(n) \leq \psi(\sqrt{2m}) \leq \sqrt{2m}\log 4 + \log^2 m.$$

From this and (7) it follows that

$$\sum\nolimits^{(1)}_{m<n\leq 2m} \Lambda(n) = \sum_{m<p\leq 2m} \log p \geq \frac{m}{3}\log 4 - 2\log^2 m - \sqrt{2m}\log 4,$$

and so

$$\sum_{m<p\leq 2m} 1 \geq \frac{1}{\log 2m}\left(\frac{m}{3}\log 4 - \sqrt{2m}\log 4 - 2\log^2 m\right).$$

If we recall that $m = [x]$, $x \geq (2n)^2$, and $x > 1024$, we easily see that the right side of the last inequality is greater than $\sqrt{x}/2 \geq n$. The lemma is proved. □

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2. Linnik's lemma

Lemma 2. *Let p be a prime number greater than n, let $1 \le P \le p^n$, and let a be an integer. Then the number T of solutions of the system of congruences*

$$\begin{aligned} &x_1 + \cdots + x_n \equiv y_1 + \cdots + y_n \pmod{p} \\ &x_1^2 + \cdots + x_n^2 \equiv y_1^2 + \cdots + y_n^2 \pmod{p^2} \\ &\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \\ &x_1^n + \cdots + x_n^n \equiv y_1^n + \cdots + y_n^n \pmod{p^n} \end{aligned}$$

$$1 - a \le x_1, \ldots, x_n, y_1, \ldots, y_n \le P - a,$$

$$x_i \not\equiv x_j \pmod{p} \quad \textit{for} \quad i \ne j,$$

satisfies the bound

$$T \le n! p^{n(n-1)/2} P^n.$$

Proof. We obviously have $T \le P^n T_1$, where T_1 is the number of solutions of the following system of congruences:

$$\begin{aligned} &x_1 + \cdots + x_n \equiv \lambda_1 \pmod{p} \\ &x_1^2 + \cdots + x_n^2 \equiv \lambda_2 \pmod{p^2} \\ &\vdots \quad \vdots \quad \vdots \quad \vdots \\ &x_1^n + \cdots + x_n^n \equiv \lambda_n \pmod{p^n} \end{aligned} \tag{8}$$

$$1 \le x_1, \ldots, x_n \le p^n,$$

$$x_i \not\equiv x_j \pmod{p} \quad \text{for} \quad i \ne j,$$

and $\lambda_1, \ldots, \lambda_n$ are a fixed n-tuple of integers. In order to estimate T_1, for each $t = 1, 2, \ldots, n$ we write x_t in the form

$$x_t = x_{1,t} + p x_{2,t} + \cdots + p^{n-1} x_{n,t},$$

where $-a + 1 \le x_{1,t} \le p - a$, $0 \le x_{2,t}, \ldots, x_{n,t} \le p - 1$. In order for an n-tuple $x_1, \ldots, x_n$ to be a solution of (8) a necessary condition is that the variables $x_{1,1}, \ldots, x_{1,n}$ satisfy the system of congruences

$$x_{1,1}^\nu + \cdots + x_{1,n}^\nu \equiv \lambda_\mu \pmod{p}, \qquad \nu = 1, 2, \ldots, n,$$

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and that the variables $x_{s,1}, \ldots, x_{s,n}$ for $s = 2, \ldots, n$, satisfy a corresponding system of linear congruences (for fixed $x_{1,1}, \ldots, x_{1,n}$):

$$x_{s,1}\left(\nu x_{1,1}^{\nu-1}\right) + \cdots + x_{s,n}\left(\nu x_{1,n}^{\nu-1}\right) \equiv \lambda_{\nu,s} \pmod{p}, \qquad \nu = s, \ldots, n,$$

where $\lambda_{s,s}, \ldots, \lambda_{n,s}$ are certain integers.

The number of solutions of the first system is at most $n!$, since it follows from the elementary theory of symmetric functions that for $p > n$ and fixed λ_ν all of the solutions of this system are permutations of a certain unique solution. Furthermore, because the variables x_t are pairwise noncongruent mod p, the coefficient matrix of each of the systems of linear congruences has maximal rank, and so the number of solutions is at most p^s. Thus, for T_1 we have the bound

$$T_1 \leq n! p \cdot p^2 \cdots p^{n-1} = n! p^{n(n-1)/2}.$$

Hence,

$$T \leq n! p^{n(n-1)/2} P^n,$$

and the lemma is proved. □

Lemma 3. *Let $\lambda_1, \ldots, \lambda_n$ be integers, let $\Lambda = (\lambda_1, \ldots, \lambda_n)$, and let $J(P; n, k; \Lambda)$ denote the number of solutions of the system of congruences*

$$\begin{aligned} x_1 + \cdots + x_{2k} &= \lambda_1 \\ \vdots \quad \vdots \quad \vdots \quad &\vdots \\ x_1^n + \cdots + x_{2k}^n &= \lambda_n, \end{aligned} \tag{9}$$

$$1 \leq x_1, \ldots, x_{2k} \leq P.$$

Then

a) $J(P; n, k; \Lambda) = \int_0^1 \cdots \int_0^1 \left|\sum_{x \leq P} \exp\left(2\pi i(\alpha_1 x + \cdots + \alpha_n x^n)\right)\right|^2 \cdot$
$\cdot \exp\left(-2\pi i(\alpha_1 \lambda_1 + \cdots + \alpha_n \lambda_n)\right) d\alpha_1 \cdots d\alpha_n;$

b) $J(P; n, k; \Lambda) \leq J(P; n, k; 0) = J(P; n, k) = J;$

c) $\sum_{\lambda_1, \ldots, \lambda_n} J(P; n, k; \Lambda) = P^{2k};$

d) $|\lambda_1| < kP, \ldots, |\lambda_n| < kP^n;$

e) $J = J(P; n, k) > (2k)^{-n} P^{2k-(n^2+n)/2};$

f) $\left|\sum_{x \leq P} \exp\left(2\pi i(\alpha_1 x + \cdots + \alpha_n x^n)\right)\right|^{2k} = \sum_{\lambda_1, \ldots, \lambda_n} J(P; n, k; \Lambda) \cdot$
$\cdot \exp\left(2\pi i(\alpha_1 \lambda_1 + \cdots + \alpha_n \lambda_n)\right).$

Proof. We obtain a) by raising the absolute value in the integrand to the $2k$-th power and integrating with respect to $\alpha_1, \ldots, \alpha_n$. Part b) follows from the fact that the absolute value of an integral does not exceed the integral of the absolute value. Part c) follows because the left side is the number of all possible $2k$-tuples $x_1, \ldots, x_{2k}$ in (9). Part d) is a consequence of the conditions on $x_1, \ldots, x_{2k}$. Part e)

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follows from c), b) and d). We obtain e) by raising the absolute value of the sum to the $2k$-th power and gathering together all terms for which $x_1+\cdots+x_{2k}=\lambda_1, \ldots, x_1^n+\cdots+x_{2k}^n=\lambda_n$. The lemma is proved. □

3. Recurrence formula for $J(P; n, k)$

We now prove a basic relation from which the mean value theorem can easily be derived.

Lemma 4. *Let $k \geq n$, $P \geq 1$. Then there exists a number p in the interval $(P^{1/n}, 2P^{1/n}]$ for which*

$$J = J(P; n, k) \leq 4k^{2n} p^{2k-2n+n(n-1)/2} P^n J(P_1; n, k-n) + (2n)^{2kn} P^k, \qquad (10)$$

where

$$P_1 = Pp^{-1} + 1.$$

Proof. If $P \leq (4n^2)^n$, then, setting $p = 2P^{1/n}$, we see that the second term in (10) is at least P^{2k}, while the first term is always nonnegative, i.e., (10) can be obtained trivially in this case. Hence, we shall suppose that

$$P > (4n^2)^n.$$

Then, by Lemma 1, the interval $(P^{1/n}, 2P^{1/n})$ contains at least n prime numbers. We let $p_1, \ldots, p_n$ denote any n distinct primes in this interval. Let

$$f(x) = \alpha_1 x + \cdots + \alpha_n x^n.$$

Then J can be represented by the multiple integral

$$J = J(P; n, k) = \int_0^1 \cdots \int_0^1 \left| \sum_{x_1 \leq P} \cdots \sum_{x_k \leq P} \exp\bigl(2\pi i \bigl(f(x_1) + \cdots + f(x_k)\bigr)\bigr) \right|^2 d\alpha_1 \cdots d\alpha_n. \qquad (11)$$

We divide the set of all k-tuples $\bar{x} = (x_1, \ldots, x_k)$ into two classes A and B, where A consists of the k-tuples $\bar{x}$ for which there exists $p_s \in \{p_1, \ldots, p_n\}$ such that there are at least n of the numbers $x_1, \ldots, x_k$ which are pairwise noncongruent mod p_s; the class B consists of all other k-tuples $\bar{x}$.

Using abbreviated notation (whose full form is obvious), we transform (11) to the following inequality:

$$J = \int_\Omega \left| \sum_{\bar{x} \in A} + \sum_{\bar{x} \in B} \right|^2 d\Omega \leq 2 \int_\Omega \left| \sum_{\bar{x} \in A} \right|^2 d\Omega + 2 \int_\Omega \left| \sum_{\bar{x} \in B} \right|^2 d\Omega = 2J_1 + 2J_2.$$

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We find a bound for J_1. J_1 is the number of solutions of the system of equations (1) for which $\overline{x} = (x_1, \ldots, x_k) \in A$ and $\overline{y} = (y_1, \ldots, y_k) \in A$. We divide the k-tuples $\overline{x} \in A$ into n sets $A_1, \ldots, A_n$, where the set A_s consists of the k-tuples corresponding to $p_s = p$, $s = 1, \ldots, n$. If a k-tuple corresponds to several p_s, then to be definite we put it in the set having the smallest p_s. Again using abbreviated notation, we have

$$J_1 = \int_\Omega \left|\sum_{\overline{x}\in A}\right|^2 d\Omega = \int_\Omega \left|\sum_{s=1}^{n}\sum_{\overline{x}\in A_s}\right|^2 d\Omega \le n\sum_{s=1}^{n}\int_\Omega \left|\sum_{\overline{x}\in A_s}\right|^2 d\Omega = n\sum_{s=1}^{n} J_{1,s}.$$

We find a bound for $J_{1,s}$. The sum inside the absolute value sign in the integral for $J_{1,s}$ has the form

$$\sum_{\overline{x}\in A_s} = \sum_{x_1}\cdots\sum_{x_k},$$

where $\overline{x} = (x_1, \ldots, x_k) \in A_s$, i.e., the numbers $x_1, \ldots, x_k$ include n which are pairwise noncongruent modulo p_s. We divide all of the k-tuples in A_s into classes (possibly overlapping) as follows. Let $t_1, \ldots, t_n$ be natural numbers, $1 \le t_1 < t_2 < \cdots < t_n \le k$. All $\overline{x} = (x_1, \ldots, x_k)$ for which $x_{t_1}, \ldots, x_{t_n}$ are pairwise noncongruent to one another mod p_s are put into the same class. Let R_1 and R_2 be two different classes in A_s. By renumbering the unknowns, we easily see that

$$\int_\Omega \left|\sum_{\overline{x}\in R_1}\right|^2 d\Omega = \int_\Omega \left|\sum_{\overline{x}\in R_2}\right|^2 d\Omega.$$

Since the total number of classes is obviously

$$\binom{k}{n} = \frac{k!}{n!(k-n)!},$$

it follows that, letting $A_{1,s}$ denote the set corresponding to $t_1 = 1, t_2 = 2, \ldots, t_n = n$, we have

$$J_{1,s} \le \binom{k}{n}^2 \int_\Omega \left|\sum_{A_{1,s}}\right|^2 d\Omega \le \binom{k}{n}^2 \int_\Omega \left|{\sum}'_{x_1,\ldots,x_n}\right|^2 \left|\sum_{x\in P} \exp\bigl(2\pi i f(x)\bigr)\right|^{2k-2n} d\Omega.$$

Here $\sum'$ means that the summation is taken over all n-tuples $x_1, \ldots, x_n$ which are pairwise noncongruent modulo $p_s = p$. We divide the second summation in the integral into p arithmetic progressions with difference p, and apply Hölder's

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inequality. We obtain

$$J_{1,s} \le \binom{k}{n}^2 \int_\Omega \left|\sum\nolimits'_{x_1,\dots,x_n}\right|^2 \left|\sum_{x\le P} \exp\bigl(2\pi i f(x)\bigr)\right|^{2k-2n} d\Omega$$

$$\le \binom{k}{n}^2 p^{2k-2n-1} \sum_{y=1}^{p} \int_\Omega \left|\sum\nolimits'_{x_1,\dots,x_n}\right|^2 \left|\sum_{0\le z\le Pp^{-1}} \exp\bigl(2\pi i f(y+pz)\bigr)\right|^{2k-2n} d\Omega.$$

Let y_0 be a value of y for which the last integral is maximal. Using the fact that whenever $x_1^0, \dots, x_k^0, y_1^0, \dots, y_k^0$ is a solution of (1), then so is $x_1^0+a, \dots, x_k^0+a$, $y_1^0+a, \dots, y_k^0+a$, we arrive at the inequality

$$J_{1,s} \le \binom{k}{n}^2 p^{2k-2n} \int_\Omega \left|\sum\nolimits''_{x_1,\dots,x_n}\right|^2 \left|\sum_{0\le z\le Pp^{-1}} \exp\bigl(2\pi i f(pz)\bigr)\right|^{2k-2n} d\Omega,$$

where the symbol $\sum''$ means that the summation is over all n-tuples of numbers $x_1, \dots, x_n$ which vary from $1-y_0$ to $P-y_0$ and are pairwise noncongruent modulo p. The integral on the right is equal to the number of solutions of the following system of equations (we let J' denote this number):

$$\begin{aligned}
&x_1+\cdots+x_n-y_1-\cdots-y_n = p(z_1+\cdots+z_{k-n}-v_1-\cdots-v_{k-n})\\
&x_1^2+\cdots+x_n^2-y_1^2-\cdots-y_n^2 = p^2(z_1^2+\cdots+z_{k-n}^2-v_1^2-\cdots-v_{k-n}^2)\\
&\vdots \qquad\quad \vdots \quad \vdots \qquad\quad \vdots \quad\quad \vdots \quad \vdots \qquad\quad \vdots\\
&x_1^n+\cdots+x_n^n-y_1^n-\cdots-y_n^n = p^n(z_1^n+\cdots+z_{k-n}^n-v_1^n-\cdots-v_{k-n}^n),
\end{aligned}$$

where the unknowns $x_1, \dots, x_n$ and $y_1, \dots, y_n$ satisfy the condition that was stated above for $x_1, \dots, x_n$, and the variables $z_1, \dots, z_{k-n}, v_1, \dots, v_{k-n}$ take all integer values from zero to Pp^{-1}.

Let $J'(Pp^{-1}; n, k-n; \Lambda)$, where $\Lambda = (\lambda_1, \dots, \lambda_n)$ is an n-tuple of integers, denote the number of solutions of the system of equations

$$z_1^\nu+\cdots+z_{k-n}^\nu-v_1^\nu-\cdots-v_{k-n}^\nu = \lambda_\nu, \qquad \nu = 1, 2, \dots, n.$$

We obviously have the inequality

$$J'(Pp^{-1}; n, k-n; \Lambda) \le J(P_1; n, k-n)$$

(in addition to the fact that J' takes its greatest value when $\Lambda = (0, \dots, 0)$, we again used the fact that adding a number to a solution of (1) gives another solution). Further let $D(\Lambda)$ denote the number of solutions of the system of equations

$$x_1^\nu+\cdots+x_n^\nu-y_1^\nu-\cdots-y_n^\nu = p^\nu\lambda_\nu, \qquad \nu = 1, 2, \dots, n.$$

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If we let $\sum_{\Lambda}$ denote the sum over all possible n-tuples Λ, we then obtain

$$J' = \sum_{\Lambda} D(\Lambda) J'(Pp^{-1}; n, k-n; \Lambda) \leq J(P_1; n, k-n) \sum_{\Lambda} D(\Lambda) =$$
$$= J(P_1; n, k-n)T,$$

where T is the number of solutions of the system of congruences

$$x_1^{\nu} + \cdots + x_n^{\nu} - y_1^{\nu} - \cdots - y_n^{\nu} \equiv 0 \pmod{p^{\nu}}, \qquad \nu = 1, 2, \ldots, n,$$

$$-y_0 + 1 \leq x_1, \ldots, x_n, y_1, \ldots, y_n \leq P - y_0.$$

We use the bound in Lemma 2 for T:

$$T \leq n! p^{n(n-1)/2} P^n.$$

As a result, if we combine the above bounds, we obtain the following inequality for J_1:

$$J_1 \leq n! \binom{k}{n}^2 n^2 p^{2k-2n+n(n-1)/2} P^n J(P_1; n, k-n),$$

where p denotes the p_s for which the right side of the last inequality is maximal.

We now find a bound for J_2. J_2 is the number of solutions of the system of equations (1) for which $\overline{x} = (x_1, \ldots, x_k) \in B$, $\overline{y} = (x_1, \ldots, x_k) \in B$. Let p_s be one of the numbers $p_1, \ldots, p_n$. For each $\overline{x} \in B$ we consider the set $\overline{x}^{(s)}$ obtained by reducing the coordinates of $\overline{x}$ modulo p_s:

$$\overline{x}^{(s)} = (x_1^{(s)}, \ldots, x_k^{(s)}), \quad x_i \equiv x_i^{(s)} \pmod{p_s}, \quad 0 \leq x_i^{(s)} < p_s, \quad i = 1, 2, \ldots, k.$$

We let B_s denote the set of k-tuples $\overline{x}^{(s)}$, and find a bound for the number of elements in B_s. This number does not exceed $\binom{p_s}{n-1}(n-1)^k$. Note that for each $\overline{x} = (x_1, \ldots, x_k) \in B$ we have the system of congruences

$$\overline{x} \equiv \overline{x}^{(s)} \pmod{p_s}, \qquad s = 1, 2, \ldots, n.$$

This system can be replaced by a single congruence of the form

$$\overline{x} \equiv \overline{M} \pmod{p_1 \cdots p_n},$$

where $\overline{M} = (M_1, \ldots, M_k)$ is a fixed k-tuple with $0 \leq M_i < p_1 \cdots p_n$, $i = 1, \ldots, k$. Since each coordinate of $\overline{x}$ is at most P, and since $P < p_1 \cdots p_n$, it follows that the last congruence is equivalent to the equation $\overline{x} = \overline{M}$. Thus, the number of k-tuples in B does not exceed the number of different $\overline{M}$, which is

$$\binom{p_1}{n-1}(n-1)^k \cdots \binom{p_n}{n-1}(n-1)^k.$$

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We now find a bound for the number of $\overline{y} = (y_1, \ldots, y_k)$ subject to the condition that they satisfy (1). If $k - n$ of them are fixed, then the others are uniquely determined up to the order of terms, i.e., there are at most $n!P^{k-n}$ different $\overline{y} = (y_1, \ldots, y_k)$. Thus,

$$J_2 \le (n-1)^{kn} \binom{p_1}{n-1} \cdots \binom{p_n}{n-1} n! P^{k-n} \le n^{2kn} P^{k-1}.$$

Combining the bounds for J_1 and J_2, we conclude that

$$J \le 2J_1 + 2J_2 \le 2n^2 \binom{k}{n}^2 n! p^{2k-2n+n(n-1)/2} P^n J(P_1; n, k-n) + (2n)^{2kn} P^{k-1}.$$

Replacing the right side by something larger, we obtain

$$J \le 4k^{2n} p^{2k-2n+n(n-1)/2} P^n J(P_1; n, k-n) + (2n)^{2kn} P^k,$$

as claimed in the lemma. □

4. Statement and proof of the mean value theorem

Theorem 1. *Let n, k and τ be natural numbers. Then for $k \ge n\tau$, $P \ge 1$ one has the bound*

$$J = J(P; n, k) \le n^{2\Delta n} 2^{\kappa} (8k)^{2n\tau} P^{2k-\Delta},$$

where

$$\Delta = \Delta(\tau) = \frac{n(n+1)}{2} - \frac{n^2}{2}\left(1 - \frac{1}{n}\right)^{\tau},$$

$$\kappa = \kappa(\tau) = n^2\tau + \frac{n(n+1)}{2}\tau - \frac{n^2(n-1)}{2}\left(1 - \left(1 - \frac{1}{n}\right)^{\tau}\right) < \frac{3(n+1)^2}{2}\tau.$$

Proof. Without loss of generality we may assume that $k = n\tau$, $n \ge 2$. We use induction on the parameter τ. The theorem holds for $\tau = 1$, since in that case $k = n$, $\Delta(1) = n$, $\kappa(1) = n^2 + n$, and the bound becomes

$$J \le n^{2n^2} 2^{n^2+n} (8n)^{2n} P^{2k-n},$$

which is somewhat worse than the simple bound

$$J \le n! P^{2k-n}.$$

We now suppose that the theorem holds for $\tau = m \ge 1$, and we prove it for $\tau = m + 1$. We apply Lemma 4 to $J(P; n, n(m+1))$, obtaining the inequality

$$J(P; n, n(m+1)) \le 4k^{2n} p^{2k-2n+n(n-1)/2} P^n J(P_1; n, k-n) + (2n)^{2kn} P^k, \quad (12)$$

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where $k = n(m+1)$. Next, we apply the theorem with $\tau = m$ to find a bound for $J(P_1; n, k-n)$:

$$J(P_1; n, k-n) \leq n^{2\Delta n} 2^{\kappa} (8nm)^{2nm} P_1^{2k-\Delta},$$

where

$$\Delta = \Delta(m) = \frac{n(n+1)}{2} - \frac{n^2}{2}\left(1 - \frac{1}{n}\right)^m,$$

$$\kappa = \kappa(m) = n^2 m + \frac{n(n+1)}{2} m - \frac{n^2(n-1)}{2}\left(1 - \left(1 - \frac{1}{n}\right)^m\right).$$

It remains to substitute this bound into (12) and show that the resulting bound is no worse than the bound in the theorem for $\tau = m+1$. Note that we may suppose that $P > (4k)^2$, since otherwise the bound in the theorem is worse than the trivial bound P^{2k}. In fact, we always have $\Delta(\tau) \leq n\tau$, and so if $P \leq (4k)^2$, then

$$P^{2k} \leq (8k)^{2n\tau} P^{2k-\Delta}.$$

Thus, $P > (4k)^2$. Then

$$pP^{-1} \leq 2P^{-1+1/n} \leq 2P^{-1/2} < (2k)^{-1};$$

hence,

$$P_1^{2k-2n-\Delta(m)} = \left(Pp^{-1} + 1\right)^{2k-2n-\Delta(m)} \leq P^{2k-2n-\Delta(m)} p^{-2k+2n+\Delta(m)} \left(1 + \frac{1}{2k}\right)^{2k}$$
$$\leq 3P^{2k-2n-\Delta(m)} p^{-2k+2n+\Delta(m)}.$$

Consequently, the first term on the right in (12) does not exceed

$$12k^{2n} n^{2\Delta(m)n} 2^{\kappa(m)} (8nm)^{2nm} p^{\Delta(m)+n(n-1)/2} P^{2k-n-\Delta(m)} \leq$$
$$\leq 12k^{2n} 2^{\kappa(m)+\Delta(m)+n(n-1)/2} n^{2\Delta(m)n} (8nm)^{2nm} \cdot$$
$$\cdot P^{2k-\left(\Delta(m)+n+1/2-\left(n/2+\Delta(m)/n\right)\right)} \leq$$
$$\leq 0.5 n^{2\Delta(m+1)n} 2^{\kappa(m+1)} (8k)^{2n(m+1)} P^{2k-\Delta(m+1)},$$

since it follows from the definition of $\Delta(\tau)$ and $\kappa(\tau)$ that

$$\Delta(m+1) = \Delta(m) + n + \frac{1}{2} - \left(\frac{n}{2} + \frac{\Delta(m)}{n}\right),$$

$$\kappa(m+1) = \kappa(m) + \Delta(m) + \frac{n(n-1)}{2}.$$

We now show that the second term in (12) is also at most

$$0.5 n^{2\Delta(m+1)n} 2^{\kappa(m+1)} (8k)^{2n(m+1)} P^{2k-\Delta(m+1)}.$$

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Since $\Delta(m+1) \leq k$, we may suppose that $P > (2n)^{2n}$; otherwise, the decrease coming from P in the bound in the theorem is less than the first factor, and the result becomes trivial. Thus,

$$P > (2n)^{2n}, \qquad \left((2n)^{-2n} P\right)^{k-\Delta(m+1)} \geq 1,$$

$$(2n)^{2kn} P^k \left((2n)^{-2n} P\right)^{k-\Delta(m+1)} \geq (2n)^{2kn} P^k,$$

i.e.,

$$(2n)^{2kn} P^k \leq n^{2\Delta(m+1)n} P^{2k-\Delta(m+1)};$$

$$0.5(2n)^{2\Delta(m+1)n} P^k \leq 0.5 n^{2\Delta(m+1)n} 2^{\kappa(m+1)} (8k)^{2n(m+1)} P^{2k-\Delta(m+1)}.$$

We have thus obtained the required bound for $J(P; n, n\tau)$ when $\tau = m+1$. The theorem is proved. □

§2. A bound for zeta sums, and some corollaries

Using the mean value theorem, one can obtain nontrivial estimates for individual trigonometric sums with a polynomial (or a function which is close to a polynomial) in the exponent. Such sums arise in the theory of the Riemann zeta-function in the following form:

$$S = \sum_{a<n\leq b} n^{it}. \tag{1}$$

A sum S of this form is called a *zeta sum.*

1. Auxiliary lemmas

We first prove two simple lemmas.

Lemma 1. *For $P > 1$ one has*

$$\left|\sum_{x=1}^{P} \exp(2\pi i \alpha x)\right| \leq \min\left(P, \frac{1}{2\|\alpha\|}\right). \tag{2}$$

Proof. Without loss of generality we may assume that $0 < \alpha < 1$. Then

$$\left|\sum_{x=1}^{P} \exp(2\pi i \alpha x)\right| = \frac{|\exp(2\pi i \alpha P) - 1|}{|\exp(2\pi i \alpha) - 1|} \leq \frac{1}{|\sin \pi\alpha|} \leq \frac{1}{2\|\alpha\|}.$$

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□

Lemma 2. *Let* $\alpha = \frac{a}{q} + \frac{\theta}{q^2}$, $q \geq 1$, $(a, q) = 1$, $|\theta| \leq 1$. *Then for any* β, $U > 0$, $P \geq 1$ *one has*

$$\sum_{x=1}^{P} \min\left(U, \|\alpha x + \beta\|^{-1}\right) \leq 6\left(Pq^{-1} + 1\right)(U + q\log q).$$

Proof. It suffices to prove that for any β_1

$$S = \sum_{x=1}^{q} \min\left(U, \|\alpha x + \beta_1\|^{-1}\right) \leq 6(U + q\log q).$$

We have the following relation:

$$\alpha x + \beta_1 = \frac{ax + [q\beta_1]}{q} + \frac{\theta_1(x)}{q^2},$$

where $\theta_1(x) = \theta x + \{q\beta_1\}q$, $|\theta_1(x)| < 2q$. Since the function $\|x\|$ is periodic with period 1, if we make the change of variables $y = ax + [q\beta_1]$ we find that

$$S \leq \sum_{|y|\leq q/2} \min\left(U, \left\|\frac{y + \theta_2(y)}{q}\right\|^{-1}\right),$$

where $|\theta_2(y)| < 2$. We may suppose that $q > 6$, since otherwise the bound for S is trivial. If $2 < |y| \leq q/2$, then

$$\left\|\frac{y + \theta_2(y)}{q}\right\| \geq \frac{|y| - 2}{q},$$

and we easily find the required bound for S:

$$S \leq 5U + \sum_{2<|y|\leq q/2} \frac{q}{|y| - 2} < 6(U + q\log q).$$

The lemma is proved. □

2. Estimate for a zeta sum

Theorem 1. *There exist two absolute constants* $c > 0$ *and* $\gamma > 0$ *such that the following bound holds for* $2 \leq N \leq t$*:*

$$\left|\sum_{n=1}^{N} n^{it}\right| \leq cN \exp\left(-\gamma \frac{\log^3 N}{\log^2 t}\right). \tag{3}$$

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Proof. Let $100 \le M \le N$. We find a bound for $|S|$, where

$$S = \sum_{n=M+1}^{2M} n^{it}.$$

Take $a = \left[M^{5/11}\right]$, $1 \le x, y \le a$. We have

$$S = \sum_{n=M+1}^{2M} \exp\bigl(it \log(n + xy)\bigr) + 2\theta a^2, \qquad |\theta| \le 1.$$

From this we obtain

$$|S| \le a^{-2} \sum_{n=M+1}^{2M} |W(n)| + 2a^2,$$

where

$$W(n) = W = \sum_{x=1}^{a} \sum_{y=1}^{a} \exp\left(it \log\left(1 + \frac{xy}{n}\right)\right).$$

We now find a bound for $|W|$. Since $|e^{i\varphi} - 1| = 2|\sin(\varphi/2)| \le |\varphi|$, it follows that for $r \ge 1$

$$\exp\left(it \log\left(1 + \frac{xy}{n}\right)\right) = \exp\bigl(it F_r(xy)\bigr) + t\theta_1 \left(\frac{a^2}{n}\right)^{r+1},$$

where

$$F_r(xy) = \sum_{m=1}^{r} \frac{(-1)^{m-1}}{m} \left(\frac{xy}{n}\right)^m, \qquad |\theta_1| \le 1.$$

We determine the integer r from the condition $r - 1 < \frac{11 \log t}{\log M} \le r$. Then

$$W = W_1 + 4\theta_2 a^2 M^{-1/13},$$

where

$$W_1 = \sum_{x=1}^{a} \sum_{y=1}^{a} \exp\left(2\pi i \left(\alpha_1 xy + \cdots + \alpha_r x^r y^r\right)\right),$$

$$\alpha_m = \frac{(-1)^{m-1}}{2\pi m} \cdot \frac{t}{n^m}, \qquad m = 1, 2, \ldots, r; \qquad |\theta_2| \le 1.$$

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Using Lemma 1.3 and Hölder's inequality, we find that for integer $k \geq 1$

$$|W_1|^{2k} \leq a^{2k-1} \sum_{x=1}^{a} \left| \sum_{y=1}^{a} \exp\left(2\pi i \left(\alpha_1 xy + \cdots + \alpha_r x^r y^r\right)\right) \right|^{2k}$$

$$\leq a^{2k-1} \sum_{\lambda_1,\ldots,\lambda_n} J(P; n, k; \Lambda) \left| \sum_{x=1}^{a} \exp\left(2\pi i \left(\alpha_1 \lambda_1 x + \cdots + \alpha_r \lambda_r x^r\right)\right) \right|.$$

Again using Hölder's inequality, Lemma 1.3 and Lemma 1, we obtain

$$|W_1|^{4k^2} \leq a^{4k^2-2k} \left(\sum_{\lambda_1,\ldots,\lambda_r} J(P; n, k; \Lambda) \right)^{2k-1} \times \sum_{\lambda_1,\ldots,\lambda_r} J(P; n, k; \Lambda) \times$$

$$\times \left| \sum_{x=1}^{a} \exp\left(2\pi i \left(\alpha_1 \lambda_1 x + \cdots + \alpha_r \lambda_r x^r\right)\right) \right|^{2k}$$

$$\leq a^{8k^2-4k} J(P; n, k) \times$$

$$\times \left| \sum_{\substack{\lambda_1,\ldots,\lambda_r \\ \mu_1,\ldots,\mu_r}} J(P; n, k, \mathbf{M}) \exp\left(2\pi i (\alpha_1 \lambda_1 \mu_1 + \cdots + \alpha_r \lambda_r \mu_r)\right) \right|$$

$$\leq a^{8k^2-4k} J(P; n, k) \sum_{\mu_1,\ldots,\mu_r} J(P; n, k; \mathbf{M}) \times$$

$$\times \min\left(2A_1, \|\alpha_1 \mu_1\|^{-1}\right) \cdots \min\left(2A_r, \|\alpha_r \mu_r\|^{-1}\right)$$

$$\leq a^{8k^2-4k} J^2(P; n, k) \prod_{m=1}^{r} \sum_{|\mu_m| < A_m} \min\left(2A_m, \|\alpha_m \mu_m\|^{-1}\right),$$

where $\mathbf{M}$ denotes the vector $(\mu_1, \ldots, \mu_r)$, and $A_m = 2ka^m$, $m = 1, \ldots, r$. We use Lemma 2 to bound the sum over μ_m for integers m in the interval

$$4\frac{\log t}{\log M} \leq m \leq 8\frac{\log t}{\log M}; \tag{4}$$

for the other m we use a trivial bound for the sum over μ_m, namely $(2A_m)^2$. For m in (4) we have

$$\sigma_m = \sum_{|\mu_m| < A_m} \min\left(2A_m, \|\alpha_m \mu_m\|^{-1}\right)$$

$$\leq 6\left(2A_m q_m^{-1} + 1\right)\left(2A_m + q_m \log q_m\right)$$

$$\leq 6(2A_m)^2 \left(q_m^{-1} + A_m^{-1} + 0.25 q_m A_m^{-2}\right) \log q_m,$$

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where

$$\alpha_m = \frac{(-1)^{m-1}t}{2\pi mn^m} = \frac{a_m}{q_m} + \frac{\theta_m}{q_m^2},$$

$$a_m = (-1)^{m-1}, \qquad q_m = \left[2\pi mn^m t^{-1}\right], \qquad |\theta_m| \le 1.$$

Using the conditions on m, we obtain

$$\sigma_m \le 400(32)^r (2A_m)^2 t^{-2/11};$$

$$|W_1|^{4k^2} \le a^{8k^2-4k} J^2(P; n, k)(400)^r (32)^{r^2} (4k)^{2r} a^{r^2+r} t^{-\frac{8}{11}\cdot\frac{\log t}{\log M}}.$$

If we choose the least integer τ for which $e^\tau \ge 380^r$, apply Theorem 1.1 to estimate $J(P; n, k)$ for $k = r\tau$, and recall that $a \le M^{5/11}$, we obtain

$$|W_1|^{4k^2} \le a^{8k^2} (400)^r (32)^{r^2} (4k)^{2r} (2r\tau)^{10r^2\tau} t^{-\frac{4}{11}\cdot\frac{\log t}{\log M}};$$

$$|W_1| \le c_1 a^2 \exp\left(-\gamma_1 \frac{\log^3 M}{\log^2 t}\right),$$

where $c_1 > 0$ and $\gamma_1 > 0$ are absolute constants. This gives us the required bound for $|S|$, and hence (3). The theorem is proved. □

Corollary 1. *For $|t| \ge 2$ one has*

$$\zeta(1 + it) = O\left(\log^{2/3} |t|\right). \tag{5}$$

This is proved directly from Theorem 1.

A bound of the form (5) for $\zeta(s)$ can also be obtained in a left neighborhood of the line $\operatorname{Re} s = 1$.

3. A bound for $\zeta(s)$ for $\operatorname{Re} s < 1$

Theorem 2. *There exists an absolute constant $\gamma_1 > 0$ such that the bound*

$$\zeta(\sigma + it) = O\left(\log^{2/3} |t|\right)$$

holds for

$$\sigma \ge 1 - \frac{\gamma_1}{\log^{2/3} |t|}, \qquad |t| \ge 2.$$

Proof. We may suppose that $\sigma \le 2$. Take $\gamma_1 = 0.5\gamma$, where $\gamma > 0$ is the absolute constant in Theorem 1, $N = \exp\left(\log^{2/3} |t|\right)$, $x = |t|$. By Corollary I.4.1 we have

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for $s = \sigma + it$:

$$\zeta(s) = \sum_{n \le N} n^{-s} + \sum_{N < n \le x} n^{-s} + O(1).$$

The absolute value of the first sum is at most

$$Y = \sum_{n \le N} n^{-\sigma} \le 1 + \int_1^N u^{-\sigma} du = 1 + \int_1^N \frac{u^{1-\sigma}}{u} du = O(\log N) = O\left(\log^{2/3} |t|\right).$$

To find a bound for the second sum, we use partial summation (Theorem A.1.1), setting

$$c(n) = n^{-it}, \qquad C(u) = \sum_{N < n \le u} n^{-it}, \qquad f(u) = u^{-\sigma},$$

and also Theorem 1:

$$\begin{aligned}
\left| \sum_{N < n \le x} n^{-s} \right| &\le \sigma \int_N^x |C(u)| u^{-1-\sigma} du + |C(x)| x^{-\sigma} \\
&= O\left(\int_N^x u^{-\sigma} \exp\left(-\gamma \frac{\log^3 u}{\log^2 |t|} \right) du \right) + O\left(|t|^{-1-\sigma-\gamma} \right) \\
&= O\left(\int_{\log N}^{\log x} \exp\left(v(1-\sigma) - \frac{\gamma v^3}{\log^2 |t|} \right) dv \right) + O(1) \\
&= O\left(\int_{\log N}^{\log x} \exp\left(-\gamma \frac{v^3}{\log^2 |t|} \right) dv \right) + O(1) = O\left(\log^{2/3} |t| \right).
\end{aligned}$$

The theorem is proved. □

From Theorem 2 it is fairly easy to obtain a refinement of de la Vallée-Poussin's theorem giving a zero-free region for $\zeta(s)$. However, first we will prove a theorem giving a bound for $|\zeta(s)|$ on the critical strip which is uniform in t and σ. A corollary of this theorem will be applied to obtain a refinement of de la Vallée-Poussin's theorem.

4. A bound for $|\zeta(s)|$ in a neighborhood of the line $\operatorname{Re} s = 1$

Theorem 3. *The following bound holds for* $0.5 \le \operatorname{Re} s = \sigma \le 1$, $|t| \ge 2$:

$$\zeta(s) = O\left(|t|^{a(1-\sigma)^{3/2}} \log |t| \right), \tag{6}$$

where $a > 0$ *is an absolute constant.*

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Proof. We suppose that $t \geq 2$. By Corollary I.4.1 with $x = t$, $s = \sigma + it$, we have

$$\zeta(s) = \sum_{n \leq t} n^{-s} + O(1).$$

If we apply partial summation to the sum over n, as in the proof of Theorem 2, we obtain

$$\begin{aligned}\left|\sum_{n \leq t} n^{-\sigma} n^{it}\right| &\leq \sigma \int_1^t |C(u)| u^{-1-\sigma}\, du + |C(t)| t^{-\sigma} \\ &= O\left(\int_1^t u^{-\sigma} \exp\left(-\gamma \frac{\log^3 u}{\log^2 t}\right) du\right) + O(1) \\ &= O\left(\int_0^{\log t} \exp\left(v(1-\sigma) - \gamma \frac{v^3}{\log^2 t}\right) dv\right) + O(1),\end{aligned}$$

where $\gamma > 0$ is the constant in Theorem 1. The integrand here reaches its maximum value when

$$v = v_0 = \sqrt{\frac{1-\sigma}{3\gamma}} \log t,$$

and this maximum value is equal to

$$\exp\left(\frac{2}{3\sqrt{3\gamma}}(1-\sigma)^{3/2} \log t\right) = t^{\frac{2(1-\sigma)^{3/2}}{3\sqrt{3\gamma}}}.$$

From this we obtain the result in the theorem with $a = 2/(3\sqrt{3\gamma})$. □

Corollary 2. *Suppose that $c > 0$ is an arbitrary fixed number, and $t \geq 10$. Then there exists a constant $c_1 > 0$ such that the bound*

$$\zeta(s) = O\left(\log^{c_1} t\right)$$

holds for $\sigma \geq 1 - c(\log\log t)^{2/3} \log^{-2/3} t$, $t \geq 10$.

The corollary follows from Theorem 3 when $\sigma \leq 1$, and from Theorem 2 when $\sigma > 1$.

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§3. Zero-free region for $\zeta(s)$

We now prove a new theorem giving a boundary for the zeros of $\zeta(s)$.

Theorem 1. *There exists an absolute constant $c > 0$ such that the Riemann zeta-function $\zeta(s)$ has no zeros in the following region of the s-plane:*

$$t \geq 10, \qquad \sigma \geq 1 - c(\log\log t)^{-1/3}\log^{-2/3} t.$$

Proof. Without loss of generality we may assume that $t \geq t_0 > 0$, where t_0 is sufficiently large. Let $\rho = \sigma + it$ be a zero of $\zeta(s)$. We set

$$\sigma = 1 - a(\log t)^{-2/3}(\log\log t)^{-1/3}, \qquad 0 \leq a \leq 1,$$

and we prove that $a \geq c_2$. Consider the point s_0 given by

$$s_0 = \sigma_0 + it, \qquad \sigma_0 = 1 + b(\log t)^{-2/3}(\log\log t)^{-1/3}, \qquad b = \frac{c}{52(c_1+1)},$$

where $c > 0$ and $c_1 > 0$ are the absolute constants in Theorem 3. We draw the circle of radius r around s_0, where

$$r = c(\log\log t)^{2/3}(\log t)^{-2/3}.$$

The point ρ is inside the circle of radius $0.5r$ centered at s_0, because $t \geq t_0$. We set $F(s) = \zeta(s)$ in Theorem A.2.5, and find a bound for $|\zeta(s)/\zeta(s_0)|$ in the disc $|s - s_0| \leq r$. By Corollary 2, we have the bound

$$\zeta(s) = O\left(\log^{c_1} t\right)$$

in this circle. In addition,

$$|\zeta(s_0)|^{-1} \leq \sum_{n=1}^{\infty} n^{-\sigma_0} \leq 1 + \int_1^{\infty} u^{-\sigma_0} du = 1 + (\sigma_0 - 1)^{-1}.$$

Hence, we have

$$\left|\frac{\zeta(s)}{\zeta(s_0)}\right| \leq M = c_2 \log^{c_1+1} t.$$

We have exactly the same bound in the disc $|s - s_1| \leq r$, where $s_1 = \sigma_0 + 2it$. Since $\zeta(s) \neq 0$ in the region $|s - s_0| \leq 0.5r$, $\mathrm{Re}(s - s_0) \geq 0$ and the region $|s - s_1| \leq 0.5r$, $\mathrm{Re}(s - s_1) \geq 0$, it follows by Theorem A.2.5 that we have

$$-\mathrm{Re}\frac{\zeta'(s_0)}{\zeta(s_0)} \leq \frac{4}{r}\log M - \mathrm{Re}\frac{1}{s_0 - \rho} \leq \left(\frac{5(c_1+1)}{\gamma_1} - \frac{1}{b+a}\right)(\log t)^{2/3}(\log\log t)^{1/3};$$

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$$-\mathrm{Re}\frac{\zeta'(s_1)}{\zeta(s_1)} \le \frac{4}{r}\log M \le \frac{5(c_1+1)}{\gamma_1}(\log t)^{2/3}(\log\log t)^{1/3};$$

$$-\frac{\zeta'(\sigma_0)}{\zeta(\sigma_0)} \le \frac{1}{\sigma_0-1} + c_3 \le \frac{1}{b}(\log t)^{2/3}(\log\log t)^{1/3}.$$

Next, using the inequality

$$0 \le 3\left(-\frac{\zeta'(\sigma_0)}{\zeta(\sigma_0)}\right) + 4\left(-\mathrm{Re}\frac{\zeta'(\sigma_0+it)}{\zeta(\sigma_0+it)}\right) + \left(-\mathrm{Re}\frac{\zeta'(\sigma_0+2it)}{\zeta(\sigma_0+2it)}\right),$$

we easily obtain

$$0 \le \frac{3}{b} - \frac{4}{b+a} + \frac{26(c_1+1)}{\gamma_1};$$

$$a \ge \frac{1 - b\cdot 26(c_1+1)\gamma_1^{-1}}{3b^{-1} + 26(c_1+1)\gamma_1^{-1}} = \frac{\gamma_1}{364(c_1+1)},$$

as was to be proved. □

Corollary. *One has the following asymptotic formulas*:

$$\psi(x) = x + O\left(x\exp\left(-c(\log x)^{0.6}(\log\log x)^{-0.2}\right)\right);$$

$$\pi(x) = \int_2^x \frac{du}{\log u} + O\left(x\exp\left(-c_1(\log x)^{0.6}(\log\log x)^{-0.2}\right)\right).$$

§4. The multidimensional Dirichlet divisor problem

The bound (6) of §2 can be applied to the multidimensional Dirichlet divisor problem to obtain an essential improvement in the results of Dirichlet, Hardy–Littlewood and Landau for large values of k.

Theorem 1. *The following asymptotic formula holds for* $x \ge 2$:

$$T_k(x) = \sum_{n\le x}\tau_k(n) = x P_{k-1}(\log x) + \theta x^{\alpha(k)}(c\log x)^k,$$

where $P_{k-1}(u)$ *is a polynomial of degree* $k-1$, $|\theta| \le 1$, $\alpha(k) = 1 - c_1 k^{-2/3}$, *and* $c > 0$ *and* $c_1 > 0$ *are absolute constants.*

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Proof. In Theorem A.5.2 we take $a(n) = \tau_k(n)$, $F(s) = \zeta^k(s)$, $A(\xi) = T_k(\xi)$, $b = 1 + \log^{-1} x$, $e^2 \le T \le x$. By Theorem A.10.3 we have

$$0 \le A(\xi) \le \xi \frac{(\log \xi + k - 1)^{k-1}}{(k-1)!},$$

and hence

$$B(b) = \int_1^\infty \frac{|A(\xi)|}{\xi^{b+1}} d\xi \le \int_1^\infty \frac{(\log \xi + k - 1)^{k-1}}{\xi^b (k-1)!} d\xi \le$$
$$\le \frac{(k-1)^{k-1}}{(k-1)!} \left(\frac{1}{b-1} + \frac{1}{(b-1)^2} + \cdots + \frac{1}{(b-1)^k} \right) \le 2(3 \log x)^k.$$

In addition,

$$\max_{x-1 \le \xi \le x+1} |A(\xi)| \le \max_{0.5x \le \xi \le 1.5x} |A(\xi)| \le 4x(3 \log x)^{k-1}.$$

Consequently, by Theorem A.5.2,

$$|R(x)| \le c \left(x^2 T^{-1} (3 \log x)^k + x(3 \log x)^{k-1} \log T \right) \le c_1 x^2 T^{-1} (3 \log x)^k.$$

We thus obtain the relation

$$\int_1^x T_k(\xi) d\xi = \frac{1}{2\pi i} \int_{b-iT}^{b+iT} \frac{\zeta^k(s)}{s(s+1)} x^{s+1} ds + c_1 \theta x^2 T^{-1} (3 \log x)^k. \tag{1}$$

We compute the last integral. By Theorem 2.3, for some fixed $a > 1$ and for $|t| \ge 2$ we have the bound

$$|\zeta(\sigma + it)| \le c|t|^{a(1-\sigma)^{3/2}} \log |t|, \qquad 1/2 \le \sigma \le 1. \tag{2}$$

We take $\alpha = 1 - (2ak)^{-2/3}$, $T = x^{1-\alpha}$, and we consider the following integral J over the rectangular contour Γ with vertices $b \pm iT$, $\alpha \pm iT$:

$$J = \frac{1}{2\pi i} \int_\Gamma \frac{\zeta^k(s)}{s(s+1)} x^{s+1} ds.$$

On the one hand, this integral is equal to the sum of the residues of the integrand. The integrand has a pole at $s = 1$ with residue equal to $x^2 P(\log x)$, where $P(u)$ is a polynomial of degree at most $k - 1$. On the other hand, J is equal to the sum of the four integrals over the sides of Γ, i.e.,

$$J = \frac{1}{2\pi i} \int_{b-iT}^{b+iT} \frac{\zeta^k(s)}{s(s+1)} x^{s+1} ds - J_1 - J_2 - J_3, \tag{3}$$

where

$$J_1 = \frac{1}{2\pi i} \int_{\alpha-iT}^{\alpha+iT} \frac{\zeta^k(s)}{s(s+1)} x^{s+1} ds,$$

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$$J_2 = \frac{1}{2\pi i}\int_{\alpha+iT}^{b+iT} \frac{\zeta^k(s)}{s(s+1)} x^{s+1} ds,$$

$$J_3 = \frac{1}{2\pi i}\int_{\alpha-iT}^{b-iT} \frac{\zeta^k(s)}{s(s+1)} x^{s+1} ds.$$

We now find a bound for the absolute value of each integral J_1, J_2, J_3. Let $4 \le T_1 \le T$; then from (2) we have

$$\left|\int_{T_1}^{2T_1} \frac{\zeta^k(\alpha+it)x^{\alpha+1+it}}{(\alpha+it)(\alpha+1+it)} dt\right| \le \frac{x^{\alpha+1}}{T_1} T_1^{ka(1-\alpha)^{3/2}} (c\log T_1)^k.$$

Since $ka(1-\alpha)^{3/2} = 1/2$, the right side of the last inequality is at most

$$T_1^{-1/2} x^{\alpha+1} (c\log x)^k,$$

and hence

$$|J_1| \le x^{\alpha+1}(c\log x)^k.$$

The integrals J_2 and J_3 are equal in absolute value. Hence,

$$|J_2| = |J_3| \le \int_\alpha^b |\zeta(\sigma+iT)|^k T^{-2} x^{\sigma+1} d\sigma \le T^{ka(1-\alpha)^{3/2}-2}(c\log T)^k x^2 \le$$
$$\le x^{\alpha+1}(c\log x)^k.$$

From the bounds for J_1, J_2 and J_3, the equality (3), and our earlier value for J, we obtain:

$$\frac{1}{2\pi i}\int_{b-iT}^{b+iT} \frac{\zeta^k(s)}{s(s+1)} x^{s+1} ds = x^2 P(\log x) + \theta x^{\alpha+1}(c\log x)^k.$$

Finally, from (1) and the last equation we find that

$$\int_1^x T_k(\xi) d\xi = x^2 P(\log x) + \theta x^{\alpha+1}(c\log x)^k.$$

Since $T_k(\xi)$ is monotonic non-decreasing as a function of ξ, it follows that for

$$h = x^{(\alpha+1)/2}$$

we have

$$T_k(x) \le \frac{1}{h}\int_x^{x+h} T_k(\xi) d\xi = \frac{1}{h}\left(\int_1^{x+h} T_k(\xi) d\xi - \int_1^x T_k(\xi) d\xi\right) =$$
$$= \frac{(x+h)^2 P\big(\log(x+h)\big) - x^2 P(\log x)}{h} + \theta h^{-1} x^{\alpha+1}(c\log x)^k =$$
$$= x P_{k-1}(\log x) + \frac{1}{2} h R_1(x,h) + \theta x^{(\alpha+1)/2}(c\log x)^k,$$

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where $R_1(x, h)$ is the second derivative of $x^2 P(\log x)$ at a point $x+\theta_1 h$, $0 \le \theta_1 \le 1$. We similarly obtain

$$T_k(x) \ge \frac{1}{h}\int_{x-h}^{x} T_k(\xi)d\xi = h^{-1}\left(\int_1^x T_k(\xi)d\xi - \int_1^{x-h} T_k(\xi)d\xi\right) =$$
$$= h^{-1}\left(x^2 P(\log x) - (x-h)^2 P\big(\log(x-h)\big)\right) + \theta_1 h^{-1} x^{\alpha+1}(c\log x)^k =$$
$$= x P_{k-1}(\log x) + 0.5h R_2(x, h) + \theta x^{0.5(\alpha+1)}(c\log x)^k,$$

where, as before, $R_2(x, h)$ is the second derivative of $x^2 P(\log x)$, but this time at a point $x - \theta_2 h$, $0 \le \theta_2 \le 1$.

It remains to bound $|R_1(x, h)|$ and $|R_2(x, h)|$ from above. In other words, we must find an upper bound for the absolute value of the residue $\Phi(x)$ of the function $\zeta^k(s)x^{s-1}$ at the point $x + \theta_3 h$, $|\theta_3| \le 1$, i.e., $|\Phi(x + \theta_3 h)|$. We have

$$\Phi(x) = \frac{1}{2\pi i}\int_{\Gamma_1} \zeta^k(s)x^{s-1}ds,$$

where Γ_1 is the rectangular contour with vertices $b \pm i$, $0.5 \pm i$, $b = 1 + \log^{-1} x$. By Corollary I.4.1 with $\operatorname{Re} s > 0$ and $N = 1$, we have

$$\zeta(s) = \frac{1}{s-1} - s\int_1^\infty \frac{\{u\}du}{u^{s+1}}.$$

The bounds $|\zeta(s)| \le c\log x$ and $|x^{s-1}| \le c$ hold on each side of Γ_1; hence,

$$|\Phi(x)| \le (c\log x)^k; \qquad |\Phi(x + \theta_3 h)| \le (c\log x)^k.$$

Using the upper and lower bounds for $T_k(x)$ and the definition of h and α, we obtain

$$T_k(x) = x P_{k-1}(\log x) + \theta x^{1-\alpha(k)}(c\log x)^k,$$

where $\alpha(k) = 0.5(2ak)^{-2/3}$. The theorem is proved. □

Remarks on Chapter IV

1. Theorem 1.1 is a variant of Vinogradov's mean value theorem. It serves as a basis for Vinogradov's method for estimating Weyl sums. For the current status of the question, see [160], [4]. Our presentation repeats that in [2].

2. The first refinement of de la Vallée-Poussin's theorem on zero-free regions for $\zeta(s)$ (and, as a corollary, an improvement of the remainder term in the prime number theorem) was obtained by Littlewood [95]. Littlewood used Weyl's method

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to estimate trigonometric sums; his result has the following form:

$$\pi(N) = \int_2^N \frac{du}{\log u} + O(R(N)), \qquad R(N) = N\exp\left(-c\sqrt{\log N \log\log N}\right).$$

An essential improvement in the results of de la Vallée-Poussin and Littlewood was obtained by Chudakov [31], using Vinogradov's method. He proved that

$$R(N) = N\exp\left(-c(\log N)^{4/7-\epsilon}\right).$$

In 1942 Vinogradov announced the following result [156], and in 1958 he published a complete proof [158] (more precisely, he published a derivation of a bound for trigonometric sums from which the result follows):

$$R(N) = N\exp\left(-c(\log N)^{3/5-\epsilon}\right).$$

Vinogradov hoped to be able to use his method to obtain an even more precise result, as is clear from the following remark in [160, p. 13]: "...It seems that a major step forward in solving the problem of the order of R (at least obtaining $R = O(N^{1-c})$, even with $c = 0.000001$) using only improved bounds for Weyl sums will be difficult without significant new advances in the theory of $\zeta(s)$."

In our exposition we followed [72].

3. The problem of finding the correct order of the remainder term in Theorem 4.1 is sometimes known as Piltz's problem. Dirichlet [38] (see also Landau [82]) proved that

$$\alpha(k) \le 1 - \frac{1}{k}, \qquad k = 2, 3, \ldots;$$

Voronoi [171] and Landau [82] proved that

$$\alpha(k) \le 1 - \frac{2}{k+1}, \qquad k = 2, 3, \ldots;$$

and Hardy and Littlewood [51] proved that

$$\alpha(k) \le 1 - \frac{3}{k+2}, \qquad k = 2, 3, \ldots.$$

In 1934, while developing his method for estimating trigonometric sums [8, p. 186], Vinogradov observed that "...for example, in the multidimensional Piltz problem

$$x_1 x_2 \cdots x_n \le a,$$

a remainder term is obtained which has order of magnitude significantly lower than any of the remainder terms obtained before."

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In 1960, Richert [137], using Vinogradov's method for estimating trigonometric sums, proved that if

$$\theta_k = \inf\left\{\xi \;\middle|\; \sum_{n\le x} \tau_k(n) - \mathrm{Res}_{s=1}\frac{x^s}{s}\zeta^k(s) = O\left(x^\xi\right)\right\},$$

then

$$\theta_k \le 1 - ck^{-2/3}.$$

In other words, $\alpha(k)$ satisfies the bound

$$\alpha(k) \le 1 - ck^{-2/3} + \epsilon.$$

Theorem 4.1 was proved in [69] (see also [68], [70]). All of the subsequent work on this theme — generalizing the problem, computing and improving the constant c_1 — follows the outline of [68] (see [41], [43], [62], [63], [103]).

Hardy [48] proved the following lower bound for $\alpha(k)$:

$$\alpha(k) \ge \frac{1}{2} - \frac{1}{2k}, \qquad k = 2, 3, \ldots.$$

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